Every ergodic interval exchange transformation is minimal

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Editorial note. The counted statement is the author's labelled lemma environment at source0.tex lines 321-323; its proof environment opens at line 325, the line after the blank separator, so the pairing is by adjacency and the retained proof range 325-337 is the whole environment.

Editorial note. The retained range stops at the end of the proof environment at line 337; the unpaired closing brace that stands at source line 338 belongs to a group opened outside the lemma and is deliberately not retained.

Editorial note. This source is new to the material-authoring work and its preamble is built from the author's own macro and theorem block (source lines 8-45) rather than from a prepared adapter.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label lem: minimality.

Editorial scope. One original result of the pinned publisher article, counted once: ergodicity implies minimality for an interval exchange transformation (author label lem: minimality).

Allowed standard prerequisites (editorial).

- Ergodicity with respect to Lebesgue measure. The transformation preserves Lebesgue measure and has no non-trivial invariant measurable set of positive measure.. It is the hypothesis, and it is used through the full measure of the set of points of continuity.
- Minimality. A transformation is minimal when the orbit of every point is dense.. It is the conclusion, and the proof establishes the equivalent epsilon formulation.
- First return map to an interval. The induced transformation on an interval J is defined by the first return time of the points of J to J.. It is the object whose maximal return exponent h and discontinuity set the proof controls.

Lemma 0.1. *If $T : I \to I$ is ergodic with respect to the Lebesgue measure then it is minimal.*

Proof. We will show that for every $x, y \in I$ and every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $|T^{-m}y - x| < 2\varepsilon$. Take an interval $J := [x, x+\varepsilon)$ and consider the first return map T_J. Let I_β^J be intervals exchanged by T_J and h_β the corresponding first return times. Since T is ergodic, the set $\tilde{I} := \bigcup_{\beta\in\mathcal{B}} \bigcup_{k=0}^{h_\beta - 1} T^k(I_\beta^J)$ is of full Lebesgue measure.

Define $h := \max\{h_\beta \mid \beta \in \mathcal{B}\}$. Consider the set

$$C := \{T^j(\partial I_\alpha) \mid \alpha \in \mathcal{A},\ j = 0, \dots, h\}.$$

Pick $0 < \delta < \varepsilon$ such that $(y, y+\delta] \cap C = \emptyset$. Since $\tilde{I}$ is of full measure, there exists $\tilde{y} \in [y, y+\delta] \cap \tilde{I}$. By the choice of δ and by the fact that T is right-continuous, the sets $\{T^{-j}[y, \tilde{y}] \mid j = 0, \dots, h\}$ are a family of pairwise disjoint intervals and T^{-1} acts on each of them by translation. Since $\tilde{y} \in \tilde{I}$, there exists $m \leq h$ such that $T^{-m}\tilde{y} \in J$ and $T^{-m}[y, \tilde{y}] = [T^m y, T^{-m}\tilde{y}]$. Thus

$$|T^{-m}y, x| < \varepsilon + \delta < 2\varepsilon,$$

which finishes the proof. □